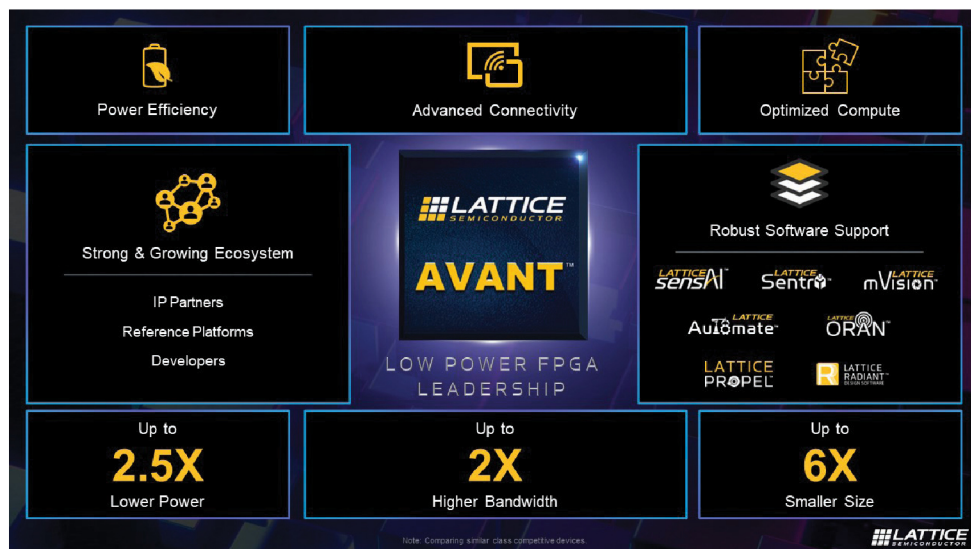


Taking Low Power to New Heights – Lattice Avant™ FPGA Platform



Addressing Key Customer Needs

Lattice Avant was born out of customer demand for compelling mid-range FPGA solutions for Communications, Computing, Industrial, and Automotive applications. When discussing their needs, it became clear that they were increasingly concerned with three key things: power efficiency, performance, and form factor – which aligned perfectly with what Lattice does best.

As compared to existing FPGAs serving the mid-range FPGA segment,

Companies today are challenged by a fast-evolving technology landscape, a seemingly insatiable hunger for more connectedness and intelligence, and an explosion of data generated at the Edge. More so than ever before, system designers and developers need efficient and flexible processing solutions that can keep pace with this accelerated demand for innovation.

These customer needs are what inspired Lattice Semiconductor to create the Lattice Avant™ platform, extending the company's low power leadership to span the small and mid-size density FPGA portions of the market.

Advancing FPGA Innovation

Architected from the ground up, Lattice Avant will enable designers with the power efficiency, performance, and small size leadership they know and expect from Lattice, with advanced connectivity and optimized compute capabilities at higher capacities than previous Lattice devices. This enables Lattice to address a broader set of customer applications than ever before.

A Closer Look at Lattice Avant

The Lattice Avant platform is designed at every level to deliver power efficiency, advanced connectivity, and optimized compute capabilities not previously available in mid-range FPGAs. While there is a lot of innovation worked into the platform, key architectural highlights include 25G SERDES and parallel I/O standards that address a wide variety of interface, and external memory interface support including DDR4, DDR5, and LPDDR4 along with legacy standards.

Lattice Avant features:

- Market-leading power efficiency with up to 2.5X lower power helping system and application engineers achieve power and thermal design efficiencies, improve operating costs, and enhance reliability
- Best-in-class performance delivering up to 2X faster throughput providing increased bandwidth and enabling reduced number of SERDES links, system cost and size.
- Industry leading form factor offering up to 6X smaller package sizes to enable size efficient system designs

Lattice Avant will leverage existing Lattice software solutions and application-specific solution stacks they're already know and understand. This familiarity will help reduce adoption time and accelerate development.

The Lattice Avant platform is architected for scalability that will enable the rapid development of multiple new device families, available today with the Lattice Avant™-E FPGA family. Lattice Avant-E FPGAs are designed to solve key customer challenges at the Edge by combining class-leading power efficiency, size and performance with an optimized feature set tailored to the needs of Edge applications like data processing and AI.

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To learn more about the Lattice Avant platform, visit the platform page: www.latticesemi.com/AvantPlatform